

CLI Lab 5 – Solutions

1. Create a working directory and navigate into it.

```
mkdir -p ~/cli_lab5 && cd ~/cli_lab5
```

2. Copy or create the file fake_syslog.txt in that folder.

```
cp /path/to/fake_syslog.txt .
```

3. Extract lines containing 'error' and redirect to errors.txt. Do the same for 'warn' into warnings.txt.

```
grep -i "error" fake_syslog.txt > errors.txt  
grep -i "warn" fake_syslog.txt > warnings.txt
```

4. Generate basic statistics for errors.txt: line count, word count, character count, longest line. Save all to summary.txt.

```
wc -l errors.txt >> summary.txt  
wc -w errors.txt >> summary.txt  
wc -c errors.txt >> summary.txt  
wc -L errors.txt >> summary.txt
```

5. Preview and save the first 5 lines of errors.txt and warnings.txt to first_errors.txt and first_warnings.txt.

```
head -n 5 errors.txt > first_errors.txt  
head -n 5 warnings.txt > first_warnings.txt
```

6. Extract timestamps (first 2 fields from each line) from errors.txt and warnings.txt. Save their frequencies to error_times.txt and warning_times.txt.

```
cut -d' ' -f1,2 errors.txt | sort | uniq -c | sort -nr > error_times.txt  
cut -d' ' -f1,2 warnings.txt | sort | uniq -c | sort -nr >  
warning_times.txt
```

7. Append the summary and timestamp statistics into a file named final_report.txt.

```
cat summary.txt > final_report.txt  
echo " >> final_report.txt  
cat error_times.txt >> final_report.txt
```

```
echo ">> final_report.txt"
cat warning_times.txt >> final_report.txt
```

8. Bonus: Try doing the timestamp count in a single pipeline (no intermediate files).

```
grep -i error fake_syslog.txt | cut -d' ' -f1,2 | sort | uniq -c | sort -nr |
head -n 5
```

9. Bonus: Identify the longest line in errors.txt using advanced CLI tools.

```
awk '{ print length, $0 }' errors.txt | sort -nr | head -n 1
```

10. Create a final report file combining all previous results (summary, error_times, warning_times).

Already done in step 7.

Expected Results & Comments

- errors.txt should contain 6 lines (with ERROR/error).
- warnings.txt should contain 4 lines (with WARNING/warning).
- Most frequent timestamp might be '2023-11-01 10:01:03' or another depending on log content.
- The longest line is likely the one with 'Unable to authenticate user'.
- Pipelines help avoid temp files and make chained operations faster.
- Useful commands: grep -i, wc -L, cut -f1,2, uniq -c, sort -nr

Reflection Questions with Solutions

11. How many lines contain 'error' and 'warn'?

There are 6 lines containing 'error' and 4 lines containing 'warn' in the fake_syslog.txt file.

12. What are the five most frequent timestamps associated with errors?

Use: grep -i error fake_syslog.txt | cut -d' ' -f1,2 | sort | uniq -c | sort -nr | head -n 5

13. What is the longest line (in characters) found in errors.txt?

Use: awk '{ print length, \$0 }' errors.txt | sort -nr | head -n 1

14. What is the difference between using > and >> in your summary file?

'>' overwrites the target file, while '>>' appends content to the existing file.

15. Which options of grep and wc did you find most useful and why?

'grep -i' for case-insensitive search, 'grep -n' for line numbers, 'wc -l' for line count, 'wc -L' for longest line length.

16. What did using a pipeline help you do more efficiently?

Pipelines let you chain multiple commands without creating intermediate files, saving time and reducing disk usage.
